Cameron Badman

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Professional Experience

Software Engineer, Gumnut.dev Sep 2025–Feb 2026

- Developed advanced Zustand and Node.js integrations for real-time collaborative state management
- Implemented Go data structures and sub-millisecond optimisations for text processing and editing pipelines
- Researched and applied garbage collection strategies in distributed computing environments
- Built presence cursor system enabling real-time multi-user awareness across collaborative sessions
- *Technologies: Go, Node.js, TypeScript, Zustand, WebSockets, Data Structures*

Software Engineer, Base Dec 2024–Sep 2025

- Implemented Github CI/CD pipelines to streamline development and deployment processes
- Managed a small group of international developers across multiple time zones
- Performed significant refactors to modularise code, enabling easy switching between monolithic and services-based architecture
- Migrated from Node PayPal SDK v0.8 to 1.1.0, improving payment processing reliability
- *Technologies: PostgreSQL, Node.js, React Native*

DevOps Engineer, Cyberloop (Oil tech) Jan 2023–Feb 2024

- Implemented CI/CD pipelines in bitbucket that reduced manual testing time by 75%
- Refactored Cypress tests and wrote majority of tests for multiple complex SaaS products
- Reduced migration costs by 20% by creating bash script to compress unstructured data on Firebase
- Developed GenAI-powered tool that assigns labels for difficulty and time automatically
- Maintained and deployed Docker containers using Portainer and Helm on Google Cloud Engine
- *Technologies: GCP, Docker, BitBucket pipelines, Jest, Playwright*

CSSE6400 Tutor, University of Queensland Feb 2026–Present

- Teaching cloud engineering concepts including Terraform, AWS, and Docker to undergraduate students
- Write assessment solutions and testing suites for course assignments

Python Tutor, Junior Engineers Jun 2021–Present

- Rewrote programming classes for children and teens, making complex coding concepts easier to understand
- Led advanced classes for 8-16 year olds, teaching Python OOP, Scratch, Arduino, Minitendo

Education

University of Queensland – Bachelors of Computer Science/Commerce 2021–Present

Projects - All projects available on GitHub

Hippocampus: *File-based vector database built for AI agent memory* [GitHub]

- Built custom O(512 log n) binary search achieving 4,200x faster exact search than FAISS at 10k vectors
- Designed file-based architecture with per-agent isolated .bin files and binary serialization (2KB per node)
- Deployed serverless infrastructure on AWS Lambda, EFS, and S3 using Terraform
- Implemented agent-to-agent orchestration with AWS Bedrock Nova for automated memory curation
- *Technologies: Go, AWS Lambda, Bedrock, EFS, S3, Terraform, Python*

IT Support AI Agent: *Self-evolving IT support agent with tiered semantic memory* [GitHub]

- Architected three siloed Hippocampus instances for company, group, and user-level policy memory
- Agent queries memories to contextualise responses, e.g. never returning Windows commands for a Mac-only organisation
- Knowledge base evolves autonomously over time as policies and preferences are learned and stored
- Built for the Claude Code Hackathon using Ollama Mistral 7B with structured tool calling
- *Technologies: Python, Go, Ollama, Hippocampus, Docker*

Gleam DynamoDB Client: *First Gleam DynamoDB client* [GitHub]

- Implemented and interfaced with the low-level AWS API integrations and documentation
- Designed type-safe monadic recursive dynamo json framework
- Learnt the gleam language in spite of documentation and llm support being non-existent
- Integrated AWS signature authentication and secure communication protocols
- *Technologies: Gleam, AWS DynamoDB, AWS SDK, Functional Programming*

Collaborative Drawing WebApp: *Real-time collaborative drawing application* [GitHub]

- Developed custom HTML canvas drawing system with real-time synchronization
- Built Go microservices with goroutines and WebSocket connections for real-time collaboration
- Deployed on AWS using containerized architecture with Kubernetes orchestration
- Implemented conflict resolution algorithms for concurrent drawing operations
- *Technologies: Go, JavaScript, Docker, Kubernetes, AWS, WebSockets*

Open-Source Contributions

goproxy: *Go HTTP proxy library forked by Stripe, Minikube, and Grafana* [GitHub]

- Developed HTTP Archive (HAR) middleware with Prometheus/Grafana integration using goroutines
- Modernizing project's testing infrastructure, transitioning to modern testing practices using testify

Community

Startup Social Brisbane – Host Meetup
IT Social Brisbane – Host Meetup

Technologies and Skills

Languages: Go, Java, Python, TypeScript, Bash, SQL, Gleam, C, C++
Web Technologies: React, Vue.js, ASP.NET, HTML, CSS
Cloud & DevOps: Amazon Web Services (AWS), Google Cloud Platform (GCP), Docker, Kubernetes, Terraform, Nix
Databases: PostgreSQL, SQLite, Redis, Cassandra, DynamoDB, Valkey
Tools: Git, GitHub, Playwright, CI/CD, Kafka, Atlassian
Operating Systems: Windows, Mac, Linux, Arch-Linux, NixOS, Ubuntu, Gentoo